

* Suppose we receive a message w/ a string of a 's & b 's: $aababababababab... \equiv \underline{X}$

Suppose a occurs w/ probability p and b w/ $1-p$

Q: How many bits of information can we extract from a long message of this kind w/ N letters

→ For large N , a occurs pN times while b occurs $(1-p)N$ times

- The number of possible messages:
 $(N! / (pN)! (1-pN)!) \sim N^N / (pN)^{pN} ((1-p)N)^{(1-p)N}$
 $= 1 / (p^p (1-p)^{(1-p)})^N =: 2^{NS}$

where, S is the Shannon entropy

$$S := -p \log(p) - (1-p) \log(1-p) \\ = - \sum_i p_i \log(p_i)$$

where we have used the Stirling formula:

* Intuition:

- (i) Our uncertainty in $\underline{X}$ before we learn the value of $\underline{X}$ or,
- (ii) a measure of how much info we have gained after we learn the value of $\underline{X}$

To each character we associate p_i
so we have a prob. dist. of the form
 $\underline{X} = (p_1, \dots, p_n)$ We thus define:

$$S(\underline{X}) := S(p_1, \dots, p_n) := - \sum_i p_i \log(p_i)$$

(Shannon Entropy)

* We take $\log(\cdot) \equiv \log_2(\cdot)$

• Note that terms for $p_i = 0$ are defined
since
 $\lim_{x \rightarrow 0} (x \log(x)) = \lim_{x \rightarrow 0} \left(\frac{\log(x)}{1/x} \right) \sim \frac{-\infty}{\infty}$
 $\lim_{x \rightarrow 0} \left(\frac{\log(x)}{1/x} \right) = \lim_{x \rightarrow 0} \left(\frac{1/x}{-1/x^2} \right) = \lim_{x \rightarrow 0} -x \rightarrow 0 \quad \checkmark$

* This def. also allows us to quantify the resources needed to store information.

Suppose we have a radio antenna (i.e. a Source.)



$X_1 X_2 \dots$

• Each are independent, identically distributed.

Q: What minimal physical resources are required to store the info being produced by the source.

A: $S(\underline{X})$ bits are required per source symbol

where $H(X) = H(X_1) = H(X_2) = \dots$ is the entropy of each random variable modeling the source.
Shannon's noiseless coding theorem

ex: Suppose an information source produces one of four symbols: x_1, x_2, x_3, x_4

- Without compression two bits of storage space corresponding to the four possible outputs are consumed for each use of the source.

$$p_1 = \frac{1}{2}, p_2 = \frac{1}{4}, p_3 = \frac{1}{8}, p_4 = \frac{1}{8}$$

→ A data compression scheme:

$$x_1: 0, x_2: 10, x_3: 110, x_4: 111$$

average length of the compressed string
 $\frac{1}{2} \cdot 1 + \frac{1}{4} \cdot 2 + \frac{1}{8} \cdot 3 + \frac{1}{8} \cdot 3 = \frac{7}{4}$ bits
of info per source.

Which is less than the 'naïve' approach of storing the source.

⇒ This matches the Entropy

$$\begin{aligned} S(X) &= -\frac{1}{2} \log\left(\frac{1}{2}\right) - \frac{1}{4} \log\left(\frac{1}{4}\right) - \frac{1}{8} \log\left(\frac{1}{8}\right) \\ &= \frac{2}{4} + \frac{1}{4} \cdot 2 + \frac{1}{4} \cdot 3 = \frac{7}{4}! \end{aligned}$$

- Any attempts to compress the source further results in data being irretrievably lost:

as the answers to fundamental questions about the physical resources required to solve some information processing problem.

→ Intuitive justification of $H(X)$

Define the information $f^n: \mathcal{I}(E)$, whose value is determined by an event E .

(i) $\mathcal{I}(E)$ is a f^n only of the prob. of the event $E \Leftrightarrow I = I(p)$

(ii) $I \in C^\infty(\mathbb{R})$

(iii) $I(pq) = I(p) + I(q)$ when $p, q > 0$

(The information gained when two ind. events occur with individual prob p, q is the sum of info gained from each source.)

Claim: $I(p) = k \log(p)$, $k \in \mathbb{R}$

$$\text{Let } I(p) = \sum_{n=1}^{\infty} \frac{f^{(n)}(0)}{n!} p^n$$

$$\Leftrightarrow n=1: p+q = pq \Rightarrow p = \frac{q}{(q-1)}$$

$$\begin{aligned} \therefore I(pq) &= \sum_{n=1}^{\infty} \frac{f^{(n)}(0)}{n!} \left(\frac{q^2}{q-1} \right)^n \\ &= \sum_{n=1}^{\infty} \frac{f^{(n)}(0)}{n!} q^n (q^2 (1 - q + \dots)^n) \\ &= \dots \end{aligned}$$

Not sure you can obtain it this way
just simply assume $I(p) = k \log(p)$

$$\Rightarrow I(p_i) = -k \log(p_i)$$

The average info gain when one of mutually exclusive set of events w/ prob. $p_1 \dots p_n$ occurs

$$S = p_1 I(p_1) + \dots = \sum_i p_i I(p_i)$$

$$S = k \sum_i p_i I(p_i)$$

Basic Properties of Entropy:

The Binary entropy:

$$S_{\text{bin}}(p) := -p \log(p) - (1-p) \log(1-p)$$

If we can plot this:

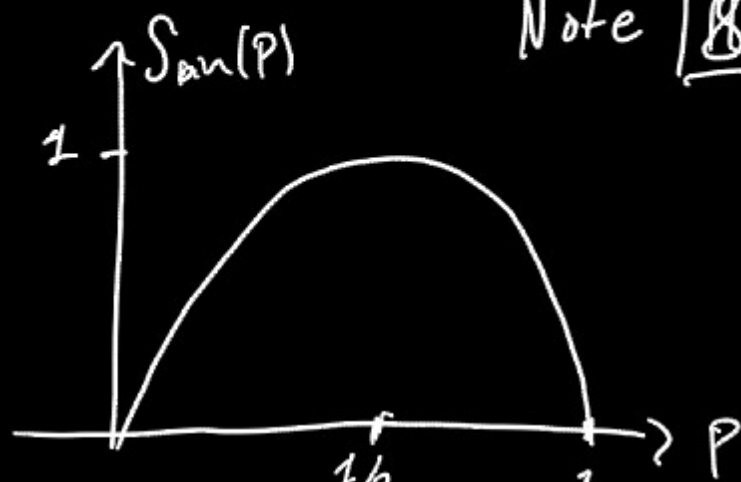
$$\begin{aligned} \frac{dS_{\text{bin}}}{dp} &= -\log(p) - 1 + \log(1-p) + 1 = 0 \\ &= \log\left(\frac{1}{p} - 1\right) = 0 \end{aligned}$$

$$\begin{aligned} \therefore 1 &= \frac{1}{p} - 1 \Leftrightarrow \boxed{p = 1/2} \text{ Maxima} \\ \frac{d^2 S_{\text{bin}}}{dp^2} &= -\frac{1}{p} - \frac{1}{1-p} \\ \frac{d^2 S_{\text{bin}}}{dp^2} \Big|_{p=1/2} &< 0 : \text{Local Maximum} \end{aligned}$$

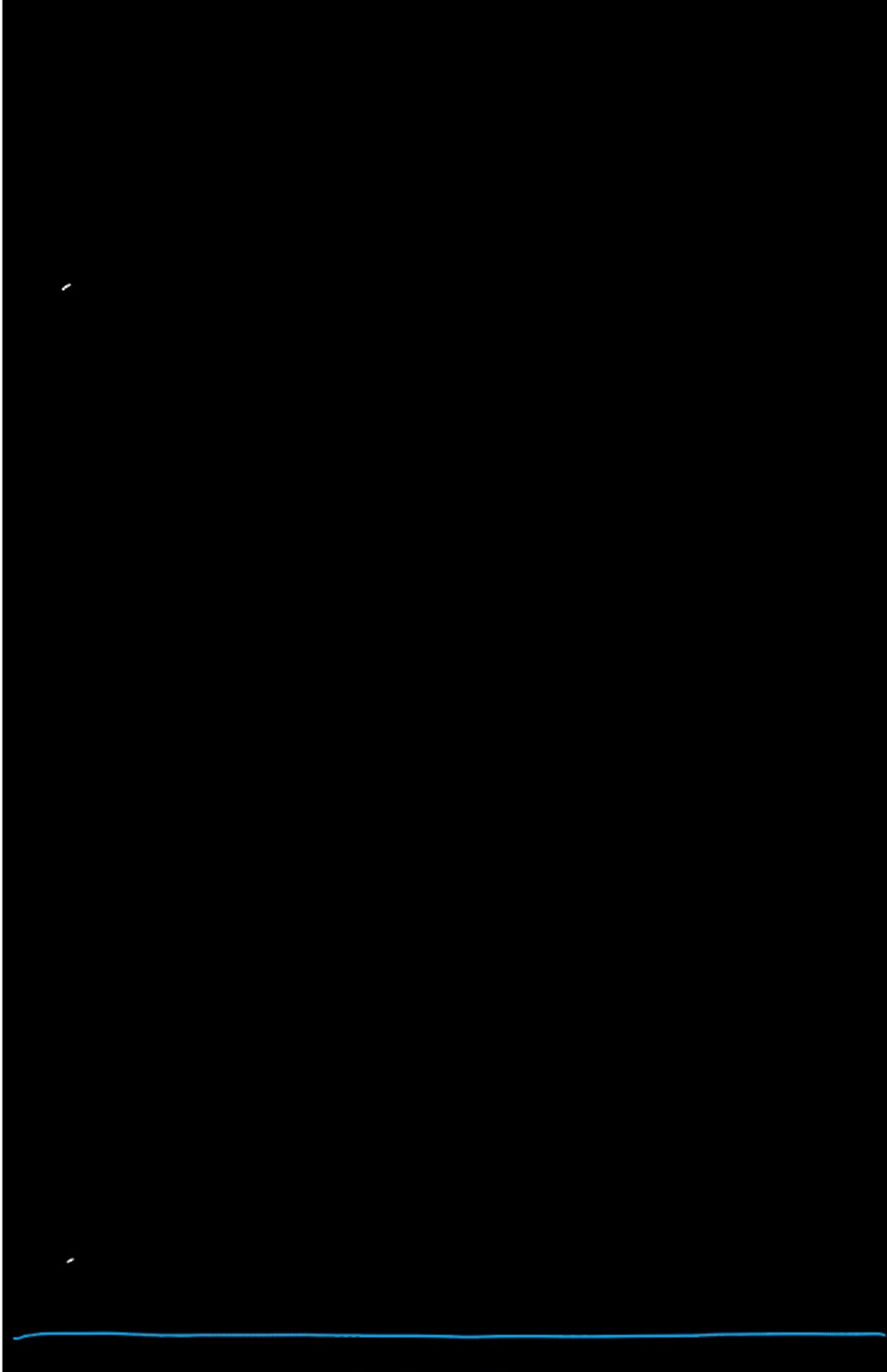
$S_{\text{bin}}(p) = 0$: Easy to see both log terms must vanish or its coeff.

$$p=0, p=1$$

$$\text{Note } \boxed{S(p) = S(1-p)}$$



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The relative Entropy:

- The relative entropy is a very useful entropy which measures the closeness of two prob. $P(x)$ and $q(x)$.
- Let $P_{X,Y}(x_i, y_j)$ be the prob. that, in a given occurrence, Alice sends $X = x_i$ and Bob hears $Y = y_j$.
The prob. that Bob hears $Y = y_j$, summing over all choices of what Alice intended is,
$$P_Y(y_j) = \sum_i P_{X,Y}(x_i, y_j)$$
- If Bob does hear $Y = y_j$, his estimate of the prob. that Alice sent x_i is the Conditional probability:
$$P_{X,Y}(x_i | y_j) := \frac{P_{X,Y}(x_i, y_j)}{P_Y(y_j)}$$

From Bob's point of view, once he has heard $Y = y_j$, his estimate of the remaining entropy in Alice's signal

$$S_{X|Y=y_j} = - \sum_i P_{X,Y}(x_i, y_j) \log(P_{X,Y}(x_i, y_j))$$

(Relative Entropy)

$$\begin{aligned} S(P(x) || q(x)) &\equiv \sum_x P(x) \log\left(\frac{P(x)}{q(x)}\right) \\ &= -S(X) - \sum_x P(x) \log(q(x)) \end{aligned}$$

Polyakov symmetries:

We must distinguish between Spacetime on M^D and the WS symm. acting on Σ .

(i) Spacetime symmetries:

We have Poincaré invariance

$$\underline{x^\mu(\sigma) \mapsto \Lambda^\mu_\nu (x^\nu - a^\nu)}, \quad \Lambda \in SO(1, d-1)$$

Recall: $(M_{\mu\nu})^\alpha_\beta := \xi^\alpha_\mu \eta_{\nu\sigma} - \delta^\sigma_\mu \eta_{\nu\alpha}$

$$\Leftrightarrow \delta x^\mu = \Lambda^\mu_\nu x^\nu + a^\mu$$

$$\delta h_{\alpha\beta} = 0$$

(local) WS symmetries:

→ Reparametrization invariance

$$\underline{\sigma \mapsto \sigma' := \sigma + \xi^\alpha(\sigma)}$$

(i) Scalar: $x^\mu(\sigma) \mapsto x'^\mu(\sigma') = x^\mu(\sigma(\sigma'))$

$$x'^\mu(\sigma') = x^\mu(\sigma) + \xi^\alpha \partial_\alpha x^\mu / \sigma$$

$$\Rightarrow \underline{\delta x^\mu = x'^\mu(\sigma) - x^\mu(\sigma) = \xi^\alpha \partial_\alpha x^\mu(\sigma)}$$

(ii) Metric:

$$\underline{\delta h_{\alpha\beta} = \nabla_\alpha \xi_\beta + \nabla_\beta \xi_\alpha}$$

(iii) Fun:

$$\delta \sqrt{-h} = \frac{1}{2} h^{\alpha\beta} (\xi^\gamma \partial_\gamma h_{\alpha\beta} + \partial_\alpha \xi^\gamma h_{\gamma\beta} + \partial_\beta \xi^\gamma h_{\alpha\gamma})$$

note $\partial_\alpha \sqrt{-h} = -\frac{1}{2} \frac{1}{\sqrt{-h}} h^{\gamma\delta} \partial_\alpha h_{\gamma\delta}$

$$\Leftrightarrow \underline{\delta \sqrt{-h} = \partial_\alpha (\xi^\alpha \sqrt{-h})}$$

* For $\Lambda^\mu_\nu = \delta^\mu_\nu + \lambda^\mu_\nu$

$$\Leftrightarrow \delta x^m = \lambda^m_{\nu} x^{\nu}$$

To see that Λ is antisymmetric:

$$\eta_{mn} x^m x^n = \eta_{mn} x'^m x'^n$$

$$\Leftrightarrow \delta(\eta_{mn} x^m x^n) = 0$$

$$2 \eta_{mn} \lambda^m_{\sigma} x^{\sigma} x^n = 0$$

$$2 \underbrace{\lambda_{n\sigma}}_{\text{Antisym}} \underbrace{x^{\sigma} x^n}_{\text{Sym}} = 0$$

examples of Poincare invariance:

(i) Rotation about x^3 by an angle θ

$$x'^1 = \cos(\theta) x^1 + \sin(\theta) x^2$$

$$x'^2 = -\sin(\theta) x^1 + \cos(\theta) x^2$$

$$\theta \ll 1, \quad \cos \rightarrow 1, \quad \sin \rightarrow \theta$$

$$x'^1 = x^1 + \theta x^2$$

$$x'^2 = -\theta x^1 + x^2$$

$$\delta x^1 = \theta x^2, \quad \delta x^2 = -\theta x^1$$

$$\text{but since } \delta x^m = a^m_{\nu} x^{\nu}$$

$$a^1_2 = \theta, \quad a^2_1 = -\theta \quad (\Rightarrow) \quad a_{12} = \eta_{11} a^1_2 = \theta$$

(ii) Boost in the x^0 & x^1 direction by φ .

$$x'^0 = \cosh(\varphi) x^0 + \sinh(\varphi) x^1$$

$$x'^1 = \sinh(\varphi) x^0 + \cosh(\varphi) x^1$$

$$\cosh \rightarrow 1, \quad \sinh \rightarrow \varphi$$

$$\delta x^0 = \varphi x^1, \quad \delta x^1 = \varphi x^0$$

$$\Rightarrow a^0_1 = \varphi, \quad a^1_0 = \varphi$$

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